Companion — RFutils user guide

This module puts a radio-mic rack on a control surface, through RFutils: battery, RF and audio state per channel across mixed vendors, and frequency programming.

The README covers installing the module. This guide is about building a page that tells you the truth about a rack.

> **Before you rely on this:** how far you can trust a *reading* depends on the vendor adapter behind it, not on this module. RFutils' README has a **Protocol status** table, and the Sennheiser adapter is labelled an *unverified skeleton*. A channel showing nothing could be the receiver, the network or the adapter.
>
> This module was built with AI assistance, directed and reviewed by a human author.

Connecting

The RFutils server, port **8420** by default.

> **No authentication, and it binds `0.0.0.0`.** Keep it on a trusted production network.

Set the battery and RF thresholds in the connection config, not on each button. That way every channel tile agrees without you editing forty buttons, and a change before doors applies everywhere.

Reading a channel tile

Colour	Means
default	fine
amber	battery low, or RF weak
red	battery critical
blue	**this channel has stopped reporting**

Blue beats everything else, and it's the one that matters.

Every threshold here goes dark when a receiver goes quiet, rather than showing its last value, because otherwise **a dead receiver looks healthy**. The variables read `--` for the same reason: a battery percentage from twenty minutes ago looks no different from a current one, and you'll only find out which it was at the worst possible moment.

Audio present and audio silent

Both take a dBFS threshold, and they answer different questions:

- `-60` catches "the mic is open".
- **Towards** `-20` catches "someone's actually talking".

Pick per page. A cue light wants the second; a fault light wants the first.

Both go dark on a stale channel, because **silence and no information aren't the same thing**, and a page that mixes them up will send someone on stage to fix a receiver that's fine.

RF is normalised across vendors

RF is reported 0–100 whatever the receiver is, so **one threshold means the same thing on every box in the rack**. That's the reason for putting a mixed rack behind one server in the first place.

Battery isn't always so obliging. Read the vendor's own units where the adapter gives you them.

Programming receivers

> **Program frequencies to receivers sends commands to real hardware** that may be in use on a show.

It's a **dry run unless *Actually transmit* is ticked**, and the log line says which happened. Leave it as a dry run while you build the page, and read the log before you tick the box.

Building a surface that fails safe

1. **Blue on every channel tile.** Without it the rack lies about its dead channels.
2. **Thresholds in the connection config**, so the whole page agrees.
3. **Audio-present at the threshold that matches the question** the page is asking.

4. **Programming buttons on their own page**, dry run by default.

Troubleshooting

Symptom	Cause
A channel is blue	It has stopped reporting. Receiver, network, or adapter — in that order of likelihood.
Variables read `--`	Same thing. A stale value is deliberately not shown.
A channel never reports anything	Check RFutils' Protocol status table for that vendor — the Sennheiser adapter is an unverified skeleton.
Audio-present never lights	The threshold is set for speech (`-20`) rather than for an open mic (`-60`).
Programming did nothing	It was a dry run. *Actually transmit* was not ticked; the log says so.

See also

- README: installing, and the full list of actions, feedbacks and variables
- `companion/HELP.md` : the same material, in Companion's help panel
- RFutils: the server, and its **Protocol status** table
